

Scale-dependent contraction of spatial wet-bulb temperature contrasts in eastern China

Jian Hou¹, Tan Meng² and Maozai Tian^{2,*}

¹College of Systems Engineering, National University of Defense Technology, Changsha 410073, China

²Center for Applied Statistics, School of Statistics, Renmin University of China, 59 Zhongguancun Street, Beijing 100872, China

*Corresponding author. mztian@ruc.edu.cn

Abstract

Regional wet-bulb temperature means omit the spatial distribution of humid heat. We compare upper-quartile and middle-half days of the monthly regional mean at 121 sites in a specified eastern-China domain. A prespecified multiscale architecture combines Gaussian-weighted semivariances at five bandwidths with equal-month, equal-scale and equal-summer aggregation. The specification was developed on the 2015 and 2022 summers and applied unchanged to 33 other summers in 1991–2025. On high-mean days, the weighted mean squared pairwise WBT contrast was 7.28% lower. This corresponds to a 3.71% reduction after square-root transformation and a 3.97% mean reduction in record-level root-mean-square pairwise differences. Contraction strengthened as the Gaussian graph bandwidth increased, from 2.86% at $h = 126$ km to 13.27% at $h = 2,013$ km. Exact Laplacian decompositions localised the broad-scale contrast to daily anomalies opposing the monthly north–south climatological pattern, while anomaly energy changed little. The pattern persisted in 1950–1990. A sparse external station comparison showed concordant broad-scale direction. High regional humid heat was associated with a flatter broad geographic field, a feature that a regional mean cannot identify.

Keywords ERA5-Land, graph Laplacian, humid heat, randomisation inference, semivariance, spatial statistics

1. Introduction

Wet-bulb temperature (WBT) is a widely used measure of humid heat stress that combines air temperature and humidity (Sherwood and Huber, 2010; Raymond et al., 2020). Two days can have identical regional means yet sharply different spatial fields: one may contain strong north–south or local contrasts, while the other is comparatively coherent. The analysis therefore focuses on spatial contrast in the WBT field itself.

Recent work has documented increasingly concentrated humid-heat extremes in standard climate regions (Speizer et al., 2022), sharp local wet-bulb extremes associated with wet soils (Chagnaud et al., 2025), and reduced intra-urban heat-stress variability during some heat waves (Shreevastava et al., 2023). Spatial-extremes models such as Healy et al. (2025) estimate nonstationary temperature tails and the area exceeding critical thresholds. The within-month comparison here examines how pairwise WBT differences change with graph scale. A regional mean discards spatial arrangement, while ordinary cross-site variance cannot distinguish neighbouring contrasts, mesoscale structure and a domain-wide gradient. Empirical variograms represent distance-dependent variation, and kernel smoothing has been used to estimate isotropic semivariograms nonparametrically (García-Soidán et al., 2004). Here, Gaussian weights define a multiscale regime-contrast summary for observed fields across the prespecified distance scales. Smooth weighting provides this summary and permits an exact spatial allocation of the regional contrast.

We ask whether high regional mean wet-bulb temperature in eastern China is accompanied by a contraction of spatial differences. The question is associational. A negative contrast is compatible with broad atmospheric forcing, but topography (Pepin et al., 2015), soil moisture (Seneviratne et al., 2010), circulation (Horton et al., 2015), and measurement support can produce or mask similar patterns. If event labels depend on spatial dispersion as well as the regional mean, the outcome influences which days are selected, and a difference can be created by construction (Simmons et al., 2011). If the most favourable distance is chosen after inspecting the data, scale-specific evidence is likewise overstated (White, 2000). Event definitions therefore use regional mean wet-bulb temperature, and the spatial scales were selected during method development.

The individual statistical ingredients used here are established. Graph Dirichlet energy is a weighted semivariance under second-order stationarity, Gaussian weighting is a standard way to smooth over pair distances, Laplacian projections are familiar, and cyclic shifts use standard randomisation logic (Cressie, 1993; Shuman et al., 2013; Dong et al., 2016). The contribution is their prespecified multiscale analysis architecture for this application: record-specific relative state contrasts, equal aggregation over months, scales and summers, additive node and structural decompositions, and a joint shift of each complete daily scale profile. Table 1 states this boundary explicitly. The architecture reveals which geographic scales and structures account for a WBT contrast without presenting any one of its statistical primitives as new.

The application uses 2015 and 2022 for method development and the other 33 summers for evaluation. A product cyclic-shift calculation assesses the multiscale statistic under a conditional product-invariance null. Equal edge splitting and Laplacian-orthogonal projections distinguish attenuation of persistent geographic structure from changes in transient anomalies. The result matters because two days with the same mean can expose quite different parts of the domain to humid heat. Broad-scale contraction may be relevant to the geographic synchrony of threshold exceedance, joint emergency demand and spatial sampling, but this study does not measure exceedance area, population exposure or health outcomes.

2. Multiscale graph regime contrasts

2.1. Scale-indexed graph dispersion and regime contrast

Let y_{it} denote WBT at location i in the selected field for day t , and let $\bar{y}_t = n^{-1} \sum_i y_{it}$. The benchmark is the site mean squared spatial deviation,

$$V_t = \frac{1}{n} \sum_{i=1}^n (y_{it} - \bar{y}_t)^2, \quad n = 121. \quad (1)$$

To retain spatial arrangement, we use an equirectangular kilometre projection centred at the mean site latitude: longitude is multiplied by $111.32 \cos(\bar{\phi})$ and latitude by 110.57. For bandwidth h , set $w_{ij,h} = \exp\{-d_{ij}^2/(2h^2)\}$ for $i \neq j$ and $w_{ii,h} = 0$. Let $\mathbf{W}_h = (w_{ij,h})$, $\mathbf{L}_h = \text{diag}(\mathbf{W}_h \mathbf{1}) - \mathbf{W}_h$, and $S_h = \sum_{i < j} w_{ij,h}$. We define graph dispersion as

$$Q_h(\mathbf{y}) = \frac{\mathbf{y}^\top \mathbf{L}_h \mathbf{y}}{2S_h} = \frac{\sum_{i < j} w_{ij,h} (y_i - y_j)^2}{2 \sum_{i < j} w_{ij,h}}. \quad (2)$$

Graph dispersion is therefore one half of the edge-weighted mean squared pairwise difference, equivalently an edge-weighted semivariance. Dividing by S_h makes the measure comparable across graphs with different total edge weights. Smaller values indicate greater graph-scale coherence. Graph dispersion measures spatial contrast, distinct from threshold-exceedance area. It has units $^\circ\text{C}^2$; $D_h^{\text{RMS}} = (2Q_h)^{1/2}$ is the weighted root mean squared pairwise WBT difference in $^\circ\text{C}$.

Equation (2) connects three familiar descriptions of spatial variation. It is the graph Dirichlet energy used to measure graph signal roughness (Shuman et al., 2013; Dong et al., 2016). If the field is second-order stationary with semivariogram $\gamma(d_{ij}) = \frac{1}{2} \text{E}(Y_i - Y_j)^2$, then

$$\text{E}\{Q_h(\mathbf{Y})\} = \frac{\sum_{i < j} w_{ij,h} \gamma(d_{ij})}{S_h}, \quad (3)$$

so its expected profile is a kernel average of the semivariogram. Under second-order stationarity, this identity supplies a semivariogram interpretation. For the nonstationary WBT fields studied here, Q_h serves as a graph-scale dispersion

measure. On a complete graph with equal off-diagonal weights, $Q_h = nV/(n-1)$. Ordinary spatial variance is therefore the complete-graph endpoint of the same family, up to its degrees-of-freedom factor.

We use five bandwidths selected during method development: $h/m \in \{0.125, 0.25, 0.5, 1, 2\}$, where m is the median pairwise distance. They correspond to 126, 252, 503, 1,006 and 2,013 km. Their weighted mean pair distances are 200, 319, 549, 826 and 992 km, and their effective edge counts are 366, 1,138, 3,223, 6,005 and 7,109 out of 7,260. Thus 126 km denotes a Gaussian bandwidth with no hard distance cut-off; it is local only relative to the sampled support, where the median nearest-neighbour distance is 150 km. For edge weights w_e , the reported Kish quantity is $N_{\text{eff}} = (\sum_e w_e)^2 / \sum_e w_e^2$.

Classifying days separately within each month-year record $r = (y, m)$ prevents the seasonal cycle or an interannual mean shift from entering the event definition. Let $q_{25,r}$ and $q_{75,r}$ be the type-7 sample quartiles of $\bar{y}_t$ in record r . A day is labelled high if $\bar{y}_t \geq q_{75,r}$ and middle if $q_{25,r} \leq \bar{y}_t < q_{75,r}$. The analysis excludes lower-quartile days. There were no threshold ties; every month-year contains eight high days and 14 or 15 middle days. These are relative upper-quartile days: the threshold is redefined within each record, and no extreme-value tail is involved.

The two groups are compared at each scale by the relative difference of their dispersion means. Let $\mathcal{E}_{ym}$ and $\mathcal{M}_{ym}$ be the high- and middle-day sets in record (y, m) , and let $\bar{Q}_{ymh}^E$ and $\bar{Q}_{ymh}^M$ denote the corresponding graph-dispersion means. The scale-specific relative contrast and its summer average are

$$R_{ymh} = \frac{\bar{Q}_{ymh}^E}{\bar{Q}_{ymh}^M} - 1, \quad \hat{\theta}_y = \frac{1}{3H} \sum_{m=1}^3 \sum_{h=1}^H R_{ymh}. \quad (4)$$

For a set $\mathcal{Y}$ of summers, the finite-record target is

$$\hat{\theta}_{\mathcal{Y}} = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \hat{\theta}_y. \quad (5)$$

A negative value means that high-regional-mean fields vary less across space. Because the bandwidths are equally spaced on the log scale, the arithmetic mean gives equal weight to every bandwidth in the prespecified five-scale grid. It is a protocol-defined finite-record summary, not a unique or scale-free regional contraction index. The raw ratio is retained as the protocol-defined primary summary. It gives a direct relative comparison but is asymmetric, can be sensitive to a small middle-day denominator, and forms a mean of record ratios rather than a ratio of pooled means. The log-ratio analysis replaces R_{ymh} by $\log(\bar{Q}_{ymh}^E/\bar{Q}_{ymh}^M)$ at the same three stages of averaging and back-transforms the final mean for interpretation. A second analysis uses the bounded symmetric contrast $2(\bar{Q}_{ymh}^E - \bar{Q}_{ymh}^M)/(\bar{Q}_{ymh}^E + \bar{Q}_{ymh}^M)$. The three contrast measures are applied to identical fields and labels in the denominator stress test. We treat the log ratio as the principal robustness analysis and the bounded symmetric form as a secondary check.

2.2. Exact spatial and structural decomposition

Graph dispersion accumulates over edges. We obtain an exact and symmetric site allocation by splitting every undirected edge equally between its endpoints, although other allocations are possible. For location i , define

$$q_{ih}(\mathbf{y}) = \frac{1}{4S_h} \sum_{j \neq i} w_{ij,h} (y_i - y_j)^2. \quad (6)$$

The identity is exact because the double sum counts every edge twice:

$$\sum_{i=1}^n q_{ih}(\mathbf{y}) = \frac{1}{2S_h} \sum_{i < j} w_{ij,h} (y_i - y_j)^2 = Q_h(\mathbf{y}). \quad (7)$$

The allocation is translation invariant and nonnegative for a single field. It also respects the graph geometry: a site receives contributions only from its incident weighted contrasts, while distant pairs carry little weight in a local graph. Equal edge splitting avoids assigning an arbitrary direction to an undirected spatial relationship.

Let $\bar{q}_{iy mh}^E$ and $\bar{q}_{iy mh}^M$ denote the high- and middle-day means within record (y, m) , and let $\bar{Q}_{ymh}^M = \sum_i \bar{q}_{iy mh}^M$. The site contribution to the record-level relative contrast is

$$c_{iy mh} = \frac{\bar{q}_{iy mh}^E - \bar{q}_{iy mh}^M}{\bar{Q}_{ymh}^M}, \quad \sum_i c_{iy mh} = R_{ymh}. \quad (8)$$

The regime-change quantity $c_{iy mh}$ may be negative. Its denominator is the record-specific middle-day dispersion. Sites in months with different baseline variability therefore enter the final allocation on the same relative scale as the corresponding record-level estimator.

Let $\mathcal{Y}$ be the set of summers. With $H = 5$, the reported node allocation is

$$c_i = \frac{1}{|\mathcal{Y}|} \sum_{y \in \mathcal{Y}} \frac{1}{3} \sum_{m=1}^3 \frac{1}{H} \sum_{h=1}^H c_{iy mh}, \quad \sum_{i=1}^n c_i = \hat{\theta}_y. \quad (9)$$

The order and weights in Equation (9) match the estimator: equal months within a summer, equal graph scales, and equal summers. The identity is checked numerically after every analysis. Changing the spatial support changes the incident edges and hence the allocation. Accordingly, every main-text map is calculated from the 121-node graph. For a scale-specific map, retaining one h in Equation (9) yields the corresponding scale-specific contrast through the observed-node sum.

For display, we fit a low-rank thin-plate REML surface to each set of 121 observed-node values and evaluate it on a fine land-masked raster. Graph construction, estimation, uncertainty calculations and node-sum identities all use the observed nodes, which are overlaid on each map. A negative c_i allocates part of the regional contraction to edges incident to site i ; the allocation is descriptive at the coordinate level. The maps also show regime means of $y_{it} - \bar{y}_t$ and their difference. These spatially centred fields separate a change in spatial pattern from a shift in the daily regional mean.

To quantify the dominant north–south structure, let ℓ be centred latitude and $\mathbf{z}_t = \mathbf{y}_t - \bar{y}_t \mathbf{1}$. Define

$$\hat{\beta}_{th} = \frac{\ell^\top \mathbf{L}_h \mathbf{z}_t}{\ell^\top \mathbf{L}_h \ell}, \quad \mathbf{e}_{th} = \mathbf{z}_t - \hat{\beta}_{th} \ell.$$

Orthogonality in the $\mathbf{L}_h$ inner product gives

$$Q_h(\mathbf{y}_t) = \frac{\hat{\beta}_{th}^2 \ell^\top \mathbf{L}_h \ell}{2S_h} + \frac{\mathbf{e}_{th}^\top \mathbf{L}_h \mathbf{e}_{th}}{2S_h}. \quad (10)$$

High-minus-middle means of the two terms, divided by middle-day total dispersion, add exactly to R_{ymh} . This exact energy accounting is descriptive. More generally, for a centred spatial basis matrix $\mathbf{B}$, the Laplacian projection has coefficients $\hat{\beta}_{th} = (\mathbf{B}^\top \mathbf{L}_h \mathbf{B})^- \mathbf{B}^\top \mathbf{L}_h \mathbf{z}_t$, where $(\cdot)^-$ is a generalised inverse. Centred latitude is the reference structural summary. Further projections add longitude and then elevation, obtained by dividing the official invariant ERA5-Land geopotential field by standard gravity. We compare the total structured and residual energies of these nested column spaces; no order-specific latitude, longitude or elevation contribution is assigned.

A second decomposition separates the persistent monthly field from daily departures. Write $\mathbf{y}_t = \boldsymbol{\mu}_m + \mathbf{a}_t$, where $\boldsymbol{\mu}_m$ is the site-specific 35-summer mean field for month m . Then

$$Q_h(\mathbf{y}_t) = \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \boldsymbol{\mu}_m}{2S_h} + \frac{\boldsymbol{\mu}_m^\top \mathbf{L}_h \mathbf{a}_t}{S_h} + \frac{\mathbf{a}_t^\top \mathbf{L}_h \mathbf{a}_t}{2S_h}. \quad (11)$$

The first term is constant within a month and cancels from the high-minus-middle numerator. The other two terms quantify change in anomaly energy and change in climatology–anomaly alignment. Dividing both differences by the middle-day total dispersion yields two components that add exactly to R_{ymh} . The cross term can be negative and is not an energy component. This is an algebraic, descriptive decomposition conditional on the chosen climatology; it does not identify circulation, soil-moisture or other physical mechanisms.

3. Finite-record inference and study design

3.1. Product cyclic-shift randomisation

Days within a month are serially dependent, so each month-year record is randomised as a unit. Its complete daily outcome profile is rotated by a randomly chosen offset while the mean-WBT series and labels remain in place. This joint rotation preserves dependence between bandwidths, and the statistic is recomputed after every shift. The one-sided Monte Carlo p -value is

$$p = \frac{1 + \sum_{b=1}^B I(T_b^{\text{shift}} \leq T_{\text{obs}})}{B + 1}. \quad (12)$$

Validity requires cyclic invariance, a stronger property than generic stationarity. For record r , let $X_r = (\bar{y}_{rt})_{t=1}^{n_r}$ denote the mean series and let D_r collect the H graph-dispersion values for each day, with $X = (X_1, \dots, X_R)$ and $D = (D_1, \dots, D_R)$. The operator C_{n_r} rotates all H columns of D_r together, and $G = \prod_r C_{n_r}$ is the group of all joint rotations. The null hypothesis is group invariance:

$$(D \mid X) \stackrel{d}{=} (gD \mid X) \quad \text{for every } g \in G. \quad (13)$$

In non-technical terms, conditional on the observed regional-mean series, independently changing the circular phase of the complete multiscale dispersion profile in any month-year record must leave the joint distribution of all profiles unchanged. Under this null, the group-rank argument gives a finite-sample-valid lower-tail test; the proof and the plus-one Monte Carlo result are in Supplementary Section S1 (Phipson and Smyth, 2010).

Truncated time-shift tests replace wrap-around invariance with strict stationarity (Harris, 2020; Yuan and Shou, 2024), but their conservative finite-sample form requires at least 19 shifts on each side for a 0.05 cutoff. Records of 30 or 31 days leave no usable aligned window under that requirement, so we use cyclic shifts under Equation (13).

The product condition is stronger than marginal cyclic invariance of each record because it permits the records to be rotated independently. It is also stronger than a null of zero mean contrast and is not a general test of no association between regional level and spatial dispersion. Common circulation or soil-moisture states across adjacent months, seasonal progression and large-scale climate drivers can violate independent phase invariance. A targeted simulation in Supplementary Section S11.4 studies one shared cross-record process, but it cannot establish the assumption for the observed climate record. Scale-wise weak family-wise error control and the development- record summaries are also detailed in Supplementary Section S1.

3.2. Development and evaluation design

The summers of 2015 and 2022 served as the design sample; the other 33 summers form the evaluation record. Project records show that these were the only complete summers used during method development, but do not document why those calendar years were first acquired. We therefore make no claim that they were randomly sampled or representative. On those summers we inspected WBT fields and graph profiles and fixed field selection, labels, spatial support, bandwidths, aggregation and quality control. Event labels, bandwidths and field selection from that stage carry into Equations (4)–(5). The target $\hat{\theta}_{\mathcal{Y}_H}$ describes the evaluation record directly; no design-summer result enters it.

For the global cyclic-shift analysis, set $T_{\text{obs}} = \hat{\theta}_{\mathcal{Y}_H}$. Each Monte Carlo transformation rotates the five-column dispersion profile within each of the 99 evaluation month-year records, then recomputes every record ratio and the equal-month, equal-scale and equal-summer mean, with $B = 99,999$.

We added the global product cyclic-shift distribution after estimating the 33-summer contrast and therefore report its p -value as exploratory. The statistic is the evaluation-record target in Equation (5). Its interpretation is restricted to the conditional product-invariance null in Equation (13).

Contrast magnitude is reported through the record mean, summer-specific contrasts, negative-summer count and leave-one-summer-out range. Supplementary Table S1 summarises the analysis sequence and the role of each calculation.

The archived Version 2 protocol states a freeze date of 2 August 2026, before the multi-year point panel was acquired or inspected. Because the first public repository commit is later, Supplementary Table S1 records the full provenance

and its evidential limit. Table 2 reconciles the protocol roles, analysis timing and final reporting. The protocol-defined Student, lag-2 Newey–West and sign-test components, and their three-component rule, remain confirmatory results in the historical record even where their final interpretation is more cautious.

3.3. Process-level reference

For comparison, a process-level reference treats the summer sequence as a stationary weakly dependent process. Under standard strong-mixing, moment and long-run variance conditions, its mean admits a central-limit approximation; a growing-bandwidth HAC estimator needs a stronger uniform autocovariance condition (Ibragimov and Linnik, 1971; Newey and West, 1987; Andrews, 1991). The full statement, ratio negative-moment conditions and proof are in the Supplement. At 33 summers, simulation shows that Student and HAC intervals can have substantial undercoverage under serial dependence. Both intervals are reported alongside the finite-record randomisation result.

4. Simulation study

The simulations examine day-level cyclic randomisation and the behaviour of process-model intervals at the 33-summer record length. The number of summers determines the repeated-summer sample size, whereas day-level simulations address within-record randomisation. Complete data-generating specifications and result tables are in Supplementary Section S11.

4.1. Data-generating processes

The day-level experiment uses the observed coordinates and six 30–31-day records. Its circular Gaussian null satisfies the invariance condition in Equation (13); two finite-autoregression variants deliberately violate wrap-around invariance. Each Gaussian anomaly is projected through $\mathbf{I} - \mathbf{1}\mathbf{1}^T/n$, so its sample mean across the 121 sites is exactly zero and the field mean equals the series used to form labels. The centred anomaly is independent of that mean. This calibration design isolates the randomisation properties; the fixed-hour and anomaly-field analyses in Section 5 examine field selection and label–field dependence in the observed record. Amplitude reduction, correlation-range expansion and gradient suppression produce true profile contractions of 5.9%–27.4%, calculated from the known quadratic forms. The Monte Carlo standard error is 0.7 percentage points at 5% rejection and 1.6 points at 50% power, so differences of one or two points are difficult to distinguish.

The comparison includes graph and binned-variogram profiles, spatial variance, two local semivariances, Moran's I and Geary's C . These are a finite set of deliberately constructed mechanisms, not a claim of general optimality. The repeated-summer experiment compares Student, fixed-lag and growing-lag calculations, the sign test and the protocol's three-component rule.

4.2. Calibration and comparison across spatial mechanisms

Figure 1 summarises the power ranges over nine spatial alternatives; complete values are in the Supplement.

Across eight cyclic-invariant designs, graph-profile rejection ranged from 4.5% to 6.8%, while the six competitors spanned 4.0%–6.4%. The graph test had rejection rates of 4.5% at temporal correlation 0.75 and 6.1% under the t_3 field. The two non-circular boundary cases gave 4.6% and 5.1%.

Gaussian-alternative absolute bias was below 0.48 percentage points and root mean squared error was 2.6–3.8 points. Under the original t_3 null, graph randomisation size was 6.1%. A paired denominator stress test then used 2,000 heavy-tailed data sets under each of the null and -7% alternatives and 999 common product shifts per data set. Null rejection was 4.70%, 4.90% and 4.75% for the raw ratio, log ratio and bounded symmetric contrast. On their native scales, the corresponding null absolute biases were 0.372, 0.070 and 0.066, and root mean squared errors were 1.472, 0.328 and 0.277. Under the -7% alternative, root mean squared errors were 1.369, 0.328 and 0.275, while rejection was 7.95%, 8.50% and 7.90%. The bounded contrast had sharply lower estimator dispersion, with similar power in this design. Randomisation calibration can coexist with unstable effect estimation; resistant summaries reduce the instability while leaving residual estimator variation.

Performance varies across the nine simulated mechanisms. Nearest-neighbour semivariance is strongest for range changes, but its power is 40.8% under weak gradient suppression, compared with 90.9% for the graph profile and

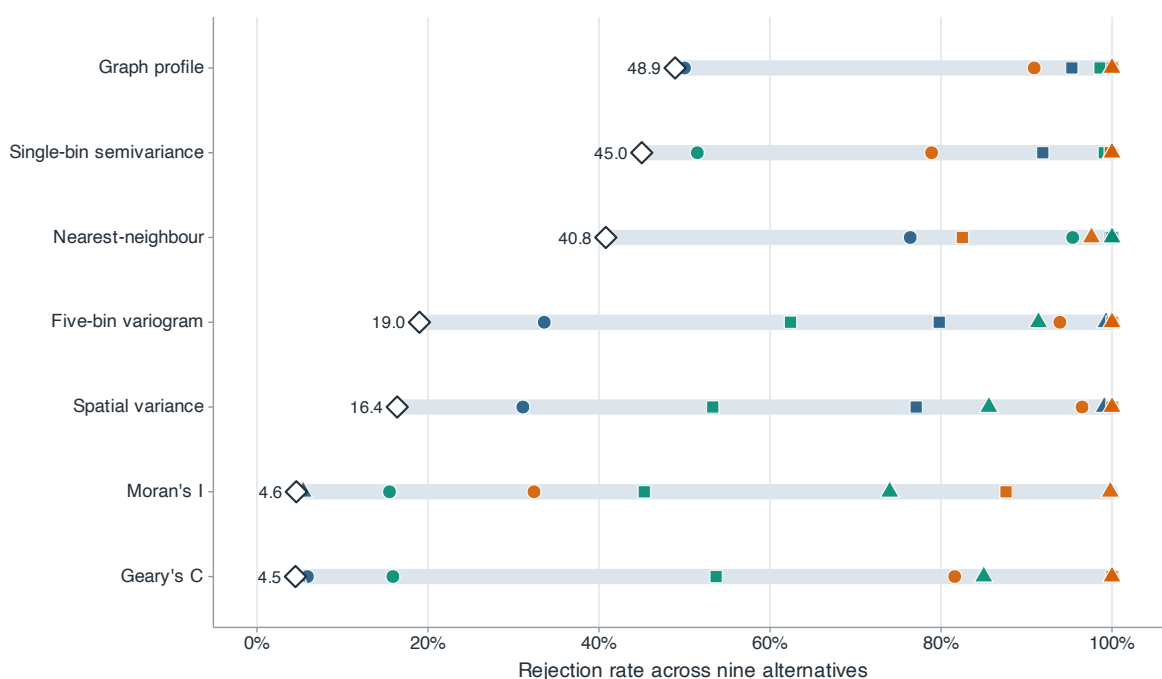


Figure 1 Power across nine simulated spatial alternatives. Blue, green and orange marks denote amplitude reduction, correlation-range expansion and gradient suppression; circles, squares and triangles denote weak, medium and strong changes. For each method, the pale segment spans power across the nine alternatives and the diamond marks the minimum.

96.5% for variance. The five-bin variogram is similar to the graph profile for gradient suppression (93.9% versus 90.9% at the weakest setting), but is less sensitive to the weak amplitude and range changes (33.6% and 19.0% versus 50.0% and 48.9%). Moran's I and Geary's C have near-nominal power for pure amplitude changes because standardisation removes amplitude. The profile's minimum power across the nine alternatives was 48.9%, higher than the minimum of each comparator within this experiment only.

The scale-specific profiles retain information lost by a scalar test. Strong amplitude reduction changes all three summaries by about 19%; range expansion acts most strongly on the local graph (28.2%), while gradient suppression acts on variance and the broad graph (43.9% and 42.5%) and changes the local graph by 9.3%.

Eight additional scenarios allow the label-generating mean and spatial field to share structure; full specifications are in the Supplement. In the shared-latent null, the sign of the spatial gradient has mean correlation 0.61 with regional WBT, but its squared energy is constant, so the true dispersion contrast is zero. The other nulls add a within-month trend, anisotropic spatial covariance or selection of a daily peak from 24 simulated hourly regional means, with hour-invariant anomaly covariance. Corresponding alternatives attenuate the gradient, impose a seasonally declining gradient or reduce high-day anomaly amplitude. Graph null rejection is 4.0%–7.0%, compared with 4.3%–7.3% for the five-bin variogram. Graph rejection under these four nulls is close to its cyclic-invariance calibration. The graph profile is at least as powerful in three of four alternatives, whereas the variogram is slightly stronger for the seasonal-gradient alternative. Smooth graph weighting and exact allocation therefore show no consistent power penalty across these scenarios.

4.3. Process-reference stress test

At $R = 33$, increasing Gaussian year correlation from 0 to 0.6 reduces Student coverage from 95.2% to 66.1%, fixed-lag coverage from 93.3% to 78.7%, and growing-HAC coverage from 91.2% to 79.0%. Corresponding Gaussian null rejection rises to 21.2%, 14.4% and 14.1%. The error arises mainly from long-run variance estimation. Coverage from all three process-model intervals deteriorates at this record length under the simulated dependence. Full results for 108 cells, including skewed and heavy-tailed innovations, appear in the Supplement.

An additional 10,000-replication experiment uses the exact 1991–2025 calendar and removes 2015 and 2022. The largest absolute mean bias over correlations 0, 0.3 and 0.6 and effects 0 and -7% is 0.03 percentage points. Excluding the two design summers therefore has negligible first-order centring bias, but short-record dependence remains: at correlation 0.6, null coverage is 68.3% for the Student interval that treats summers as independent and 80.2% for calendar lag 2.

5. Humid-heat application

We apply graph dispersion to synchronous wet-bulb temperature fields in eastern China during 1991–2025 and extend the analysis to 1950–1990.

5.1. Data construction and analysis design

ERA5-Land supplies hourly land-surface fields on a 0.1° latitude–longitude grid ([Muñoz-Sabater et al., 2021](#)). We use 2-m air temperature, 2-m dew-point temperature and surface pressure for June–August 1991–2025 over $105\text{--}125^\circ\text{E}$ and $20\text{--}42^\circ\text{N}$. The rectangle provides a regular support spanning the subtropical-to-temperate north–south gradient of eastern China while excluding most of the Tibetan Plateau west of 105°E . Its northern and southern limits bound that gradient. The analysis panel begins in 1991.

The historical extension covers all 41 June–August seasons from 1950, the first year of the ERA5-Land record, through 1990. It uses the same sites, field construction and five-scale estimator. We report separate summaries for 1950–1978 and 1979–1990 because the earlier segment uses the preliminary ERA5 back extension ([ECMWF production documentation](#)). The extension plan and its 99,999-draw product cyclic-shift calculation were recorded before the historical values were retrieved.

Every 16th longitude and 18th latitude cell from a fixed north-west origin gives a 13×13 candidate lattice; removing ocean cells leaves 121 sites (Figure 2). The 35 summers contain 9,350,880 quality-controlled hour–site records and 92 complete UTC fields per summer. Because of NetCDF packing, 2,092 dew-point values exceeded air temperature slightly (0.022% of records); these pairs were clipped at equality before WBT was calculated. The main target is a site average. For area weighting, $a_i = \cos(\text{latitude}_i)$,

$$\bar{y}_t^{(a)} = \frac{\sum_i a_i y_{it}}{\sum_i a_i}, \quad Q_{h,a}(\mathbf{y}) = \frac{\sum_{i < j} a_i a_j w_{ij,h} (y_i - y_j)^2}{2 \sum_{i < j} a_i a_j w_{ij,h}}.$$

The first area-weighted calculation replaces Q_h by $Q_{h,a}$ using the original labels; the second redefines the labels from $\bar{y}_t^{(a)}$ at the regional peak hours. The two calculations separate spatial weighting from event relabelling.

WBT is calculated from temperature T and dew point T_d in kelvin and surface pressure p in pascals. Vapour pressure is $e(T_d) = 611.2 \exp\{17.67(T_d - 273.15)/(T_d - 29.65)\}$ Pa, and the Bolton mixing-ratio, lifting-condensation-temperature and equivalent-potential-temperature relations are evaluated in these units ([Bolton, 1980](#)). Saturated T_w is obtained by solving

$$\theta_e(T_w, T_w, p) = \theta_e(T, T_d, p) \quad (14)$$

at the original pressure with 40 bisection iterations on $[T_d, T]$; even a 100-K initial interval then falls below 10^{-9} K. Values with $T_d > T$ are clipped at equality before evaluation. The Stull approximation provides a comparison ([Stull, 2011](#)); the main calculation follows the pseudoadiabatic definition discussed by [Davies-Jones \(2008\)](#).

For each UTC day, the 121 simultaneous values at the peak regional mean form the daily field. Within each month-year, the upper regional-mean quartile is high, the middle half is middle and the lower quartile is excluded. Labels depend on regional mean WBT, and the selected hour maximises that mean; graph dispersion enters after selection. UTC+8 days, alternative thresholds, sitewise maxima, all 24 fixed UTC hours and daily-mean fields provide comparisons. At each fixed hour, both the original peak-based labels and labels recomputed from that hour's regional mean are analysed; daily-mean labels are likewise recomputed from daily-mean WBT. A dated event manifest records the type-7 thresholds, membership and absence of ties.

Three checks address within-month seasonal progression: comparison of the day-of-month distributions, removal of a separate linear day-of-month trend at every site within each year-month, and subtraction of each site's calendar-day climatology, estimated from all summers except the one being analysed. The last two calculations use the original

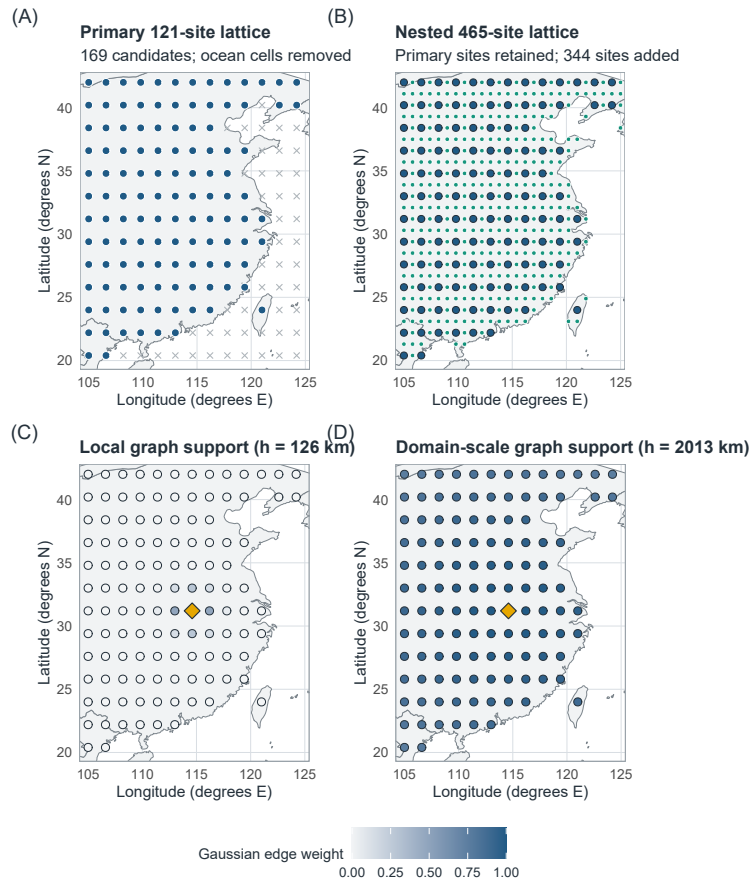


Figure 2 Spatial sampling and graph support. (A) Analysis lattice with 121 land sites from 169 candidates. (B) Nested lattice adding 344 land sites while retaining every analysis site. (C,D) Gaussian edge weights from one reference site under the 126- and 2,013-km graphs, showing how bandwidth changes the geographic support at fixed coordinates. Basemap: Natural Earth 1:50m, accessed through the R maps package.

event labels. A date-matched calculation compares each high day with all middle days in the same month-year lying within ± 3 or ± 5 calendar days. It first averages the available high-day-specific ratios and then uses the same equal-month, equal-scale and equal-summer hierarchy as the estimator; unmatched-day and empty-record counts are retained.

A continuous diagnostic regresses $\log Q_h$ on centred regional mean WBT within each month-year and bandwidth. Slopes are then averaged equally over months, bandwidths and evaluation summers. This slope supplements the quartile target with a graded summary.

Alternative graph kernels are $\exp(-d_{ij}/\rho)$ and $[1 - d_{ij}/\rho]_+^2$. For each Gaussian scale, ρ is found by bisection so that the alternative kernel has the same edge-weighted mean pair distance. Across the five scales, their effective edge counts range from 334 to 7,112 and from 382 to 7,113, compared with 366 to 7,109 for the Gaussian graphs; the full matched-distance and effective-edge metadata accompany the analysis.

A nested $0.8^\circ \times 0.9^\circ$ lattice adds 344 land sites while retaining all 121 locations. Numerical convergence is assessed at the five physical bandwidths by comparing 121 sites with a 239-site longitude refinement, a 236-site latitude refinement and all 465 sites under the evaluation events; peak times and labels are also reselected on the full grid. Thin-plate REML surfaces (Wood, 2003) are used only for display in Figures 3 and 4; every estimate uses observed nodes.

Additional spatial-support checks use WGS84 geodesic distances on the 121 sites and the 465-site lattice. On the latter, each boundary is moved inward by one or two lattice steps and event labels are recomputed from the retained domain. A non-rectangular support intersects the downloaded grid with the Natural Earth low-resolution China feature; Taiwan is a separate feature in that source and is not included. Only inward changes are estimable because no fields were downloaded outside the original rectangle. These checks retain the five physical bandwidths and report complete scale curves.

We compare ERA5-Land WBT with observations from the National Oceanic and Atmospheric Administration Integrated Surface Database (NOAA ISD) (Smith et al., 2011). The station comparison uses ten summers (1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2023 and 2025) selected before station retrieval. Archive, operating-period and geographic eligibility rules followed by a deterministic maximin design selected 30 dispersed stations. A station-year was retained when it had at least 400 quality-controlled, complete exact-hour observations of temperature, dew point and sea-level pressure; station elevation supplied surface pressure under a standard atmosphere, $p_s = p_{SL} \{1 - 2.25577 \times 10^{-5} z\}^{5.2559}$ for elevation z in metres. At two offshore stations, all variables were missing from the nearest ERA5-Land grid cells, leaving 28 usable stations and 175,172 exact-hour WBT pairs. The station-field calculation uses peak times and high-middle labels from the 121-site ERA5-Land fields. At each retained time, NOAA and ERA5-Land graph contrasts use the identical station subset and require at least 10 stations. All 30 year-month records retained at least one field in each regime. The 503-, 1,006- and 2,013-km graphs were designated as station-supported before evaluation; the two narrower graphs diagnose the limit of the station geometry. The comparison assesses measurement agreement and the direction of spatial contrast on this sparse network. Because times and labels come from ERA5-Land and station support varies across fields, it is not an independent event replication.

5.2. Evaluation-period multiscale result

Across the 33 evaluation summers, the five-scale contrast on the weighted squared-difference scale was -7.28% . The protocol-defined three-component consistency rule was satisfied. The one-sided Student and lag-2 Newey–West reference values were 2.56×10^{-6} and 3.54×10^{-6} ; the independent-summer exact binomial sign reference value was 3.31×10^{-5} . All three were below the prespecified 0.025 threshold. The corresponding two-sided Student and lag-2 HAC reference intervals were $[-10.00, -4.57]\%$ and $[-10.05, -4.51]\%$. Subsequent simulations nevertheless showed that Student and fixed-lag HAC intervals can under-cover under appreciable interannual dependence. We report the protocol decision but do not treat it as definitive long-run process inference.

In the post-result exploratory calculation, no sampled shift was as negative as the observed statistic. With $B = 99,999$ draws, an exceedance count of zero and seed 20260809, the plus-one Monte Carlo value was $p_{MC} = 1.0 \times 10^{-5}$, the minimum attainable value for this run. The complete five-scale daily profile was shifted jointly within each of the 99 month-year records. This result applies only under the conditional product-invariance null in Equation (13).

Twenty-eight summer-specific contrasts were negative, and leave-one-summer-out estimates ranged from -7.69% to -6.64% . The Student interval above is a descriptive t -reference interval based on between-summer variation, not a confidence interval under serial dependence. The fraction negative is 0.85, with a binomial Wilson reference interval of $[0.69, 0.93]$ under independent summer signs and a one-sided independent-summer sign reference value of 3.3×10^{-5} .

The fitted trend was -0.22 percentage points per year (95% descriptive OLS interval based on between-summer variation, $[-0.48, 0.05]$, two-sided $p = 0.10$). The 2008–2025 mean was 2.95 points more negative than the 1991–2007 mean, with a descriptive Welch interval based on between-summer variation of $[-8.43, 2.52]$ points. Both intervals include zero, leaving the temporal trend uncertain.

Contrasts increase in magnitude from -2.86% as the Gaussian bandwidth is $h = 126$ km to -13.27% at $h = 2,013$ km (Table 3; Figure 3). Negative-summer counts rise from 22 to 30. A 31-point log-spaced bandwidth profile decreases at each successive evaluated bandwidth across the same $h = 126$ – $2,013$ -km range. Complete-graph variance falls 14.45% and is negative in all 33 summers.

Historical extension, 1950–1990

The 1950–1990 extension gave a five-scale contrast of -11.04% (descriptive t -reference interval based on between-summer variation, $[-12.84, -9.24]\%$). Forty of 41 summer contrasts were negative, and the leave-one-summer-out

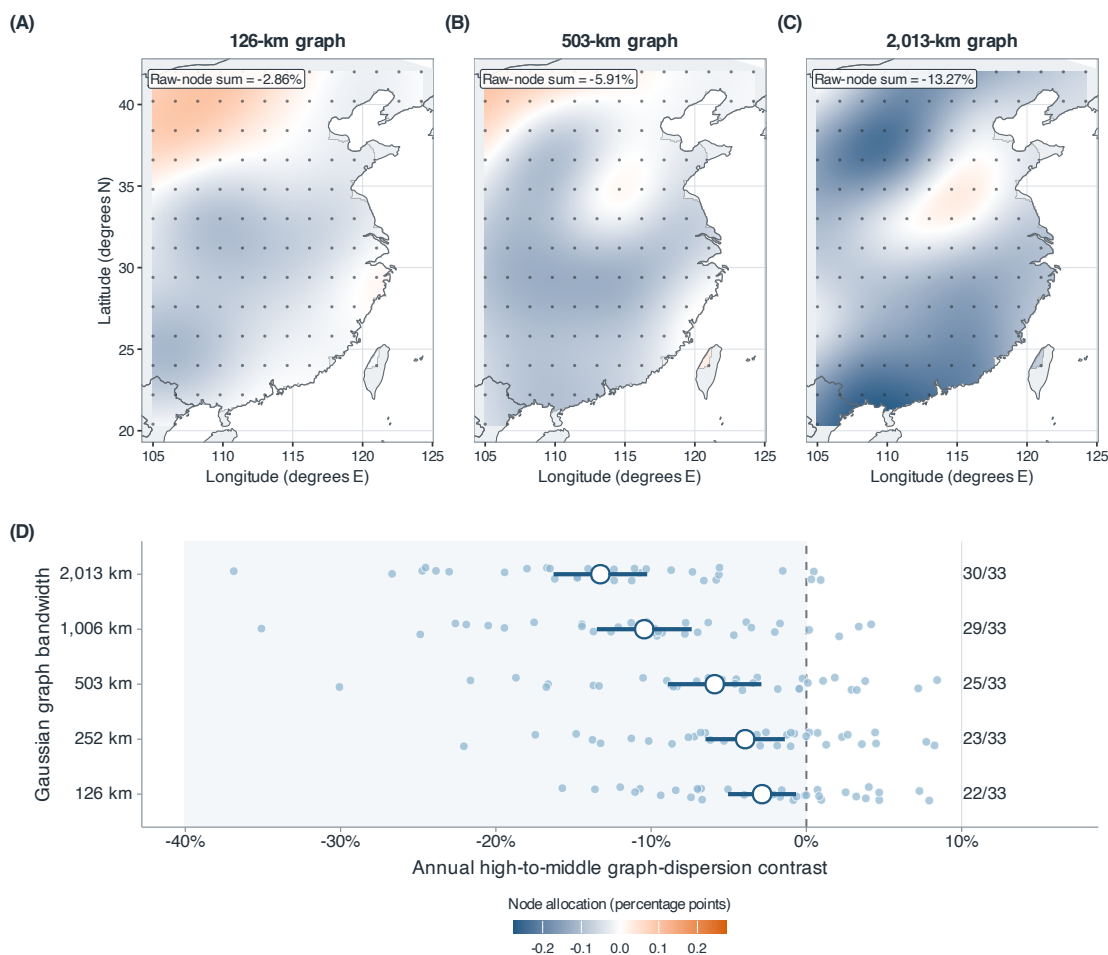


Figure 3 Scale-specific node allocations and summer contrasts for the 33 evaluation summers. (A–C) Allocations for the 126-, 503- and 2,013-km graphs share a symmetric colour scale; their observed-node sums are -2.86% , -5.91% and -13.27% . The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes. (D) Small circles show summer-specific contrasts at all five bandwidths; large circles and bars show means and 95% descriptive (t)-reference intervals based on between-summer variation. Labels give the numbers of negative summers.

range was -11.33% to -10.62% . Its 99,999-draw product-shift calculation produced no statistic as negative as the observed value (plus-one $p_{MC} = 1.0 \times 10^{-5}$, the minimum attainable value for that run, under its conditional product-invariance null). The scale profile was -5.96 , -7.51 , -10.08 , -14.50 and -17.17% from 126 to 2,013 km, with 38, 39, 40, 41 and 41 negative summers. The negative sign and strengthening contraction with bandwidth also appear in the 41 earlier summers.

The production-history split gave -10.50% in 1950–1978 (28 of 29 summers negative) and -12.34% in 1979–1990 (12 of 12 negative). The segments are reported separately because the pre-1979 record uses different forcing. The historical climatology–anomaly decomposition re-estimates the site-specific monthly climatological field from the 41 summers in 1950–1990. At 2,013 km, the historical anomaly–energy component was -0.28 points (95% descriptive (t)-reference interval based on between-summer variation, $[-1.01, 0.45]$), whereas the climatology–anomaly cross component was -16.89 points ($[-19.31, -14.47]$) and negative in all 41 summers. Both the latitude and latitude–longitude structured components were negative in every summer at every bandwidth. In the earlier record, most

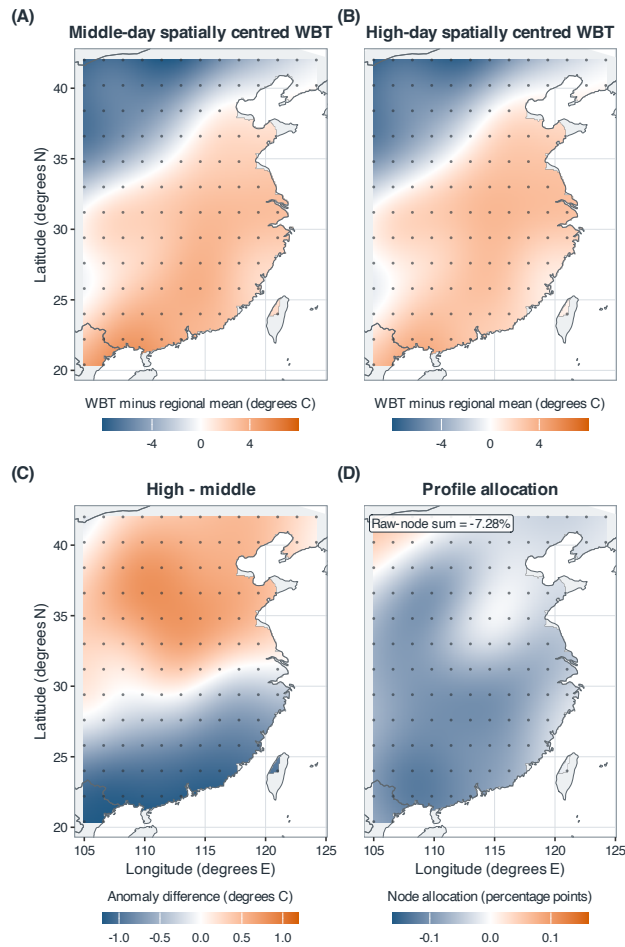


Figure 4 Spatial structure of the 121-site evaluation contrast. (A,B) Middle- and high-day spatially centred fields; (C) their difference; (D) exact observed-node allocations averaged over the five graph scales. The 121 allocations sum to -7.28% , identical to the five-scale estimand. The interpolated surfaces visualise the 121 observed-node values; estimation and inference use the observed nodes.

broad-scale contraction was likewise in the alignment term. The 1950–1978 estimates inherit the uncertainty of the preliminary forcing data.

5.3. Spatial structure and energy decomposition

The decomposition in Equation (10) localises most of the contrast to the north–south gradient. At 126 km it accounts for -2.58 points (95% (t)-reference interval under an independent-summer model, $[-2.92, -2.24]$) of the -2.86 -point contrast. From 252 to 2,013 km its mean contribution is more negative than the total ($-4.60, -8.38, -12.73, -14.96$ points), while residual contrasts contribute $+0.66, +2.48, +2.29, +1.69$ points. The gradient component is negative in all 33 evaluation summers at every bandwidth. Adding longitude to the basis makes the 2,013-km structured component -16.23 points, compared with -14.96 points for latitude alone; the corresponding residuals are positive. The latitude–longitude–elevation basis gives a 2,013-km structured component of -14.34 points (95% descriptive (t)-reference interval based on between-summer variation, $[-16.75, -11.92]$), negative in all 33 summers, and a residual of $+1.07$ points. At 126 km, however, its structured component is -1.16 points (95% descriptive (t)-reference interval based on between-summer variation, $[-2.66, 0.33]$) and is negative in only 18 summers. At broad scales, the nested bases show that latitude accounts for most of the low-dimensional

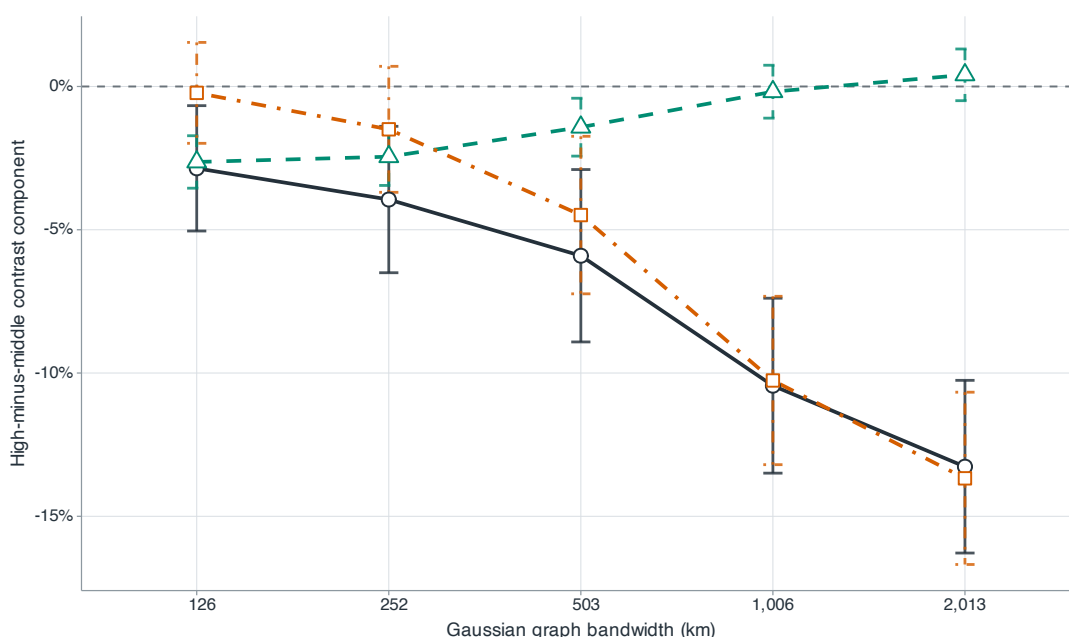


Figure 5 Exact decomposition of the raw-field graph contrast. Dark circles, teal triangles and orange squares denote total change, anomaly-energy change and the climatology-anomaly cross component; bars are 95% descriptive intervals. Monthly-climatology energy cancels within each record.

geographic-topographic structure; structured localisation is weaker at 126 km. Node allocations are negative at 106 sites and sum exactly to -7.28% . Each is a symmetric incident-edge allocation.

The climatology-anomaly decomposition in Equation (11) provides a descriptive location of the broad-scale change. At 126 km, anomaly-energy change contributes -2.63 points and the climatology-anomaly cross term contributes -0.22 points. At 2,013 km, the anomaly-energy component is $+0.41$ points (95% descriptive (t)-reference interval based on between-summer variation, $[-0.50, 1.31]$), whereas the cross term is -13.68 points (95% descriptive (t)-reference interval based on between-summer variation, $[-16.68, -10.67]$) and is negative in all 33 summers. Thus the broad-scale contraction was accounted for algebraically by high-day anomalies opposing the chosen monthly climatological pattern, while their own broad-scale energy changed little. The cross term is not an energy and this accounting does not identify a physical mechanism.

For each evaluation summer, a sensitivity calculation rebuilt the three monthly climatological fields from the other 34 summers and repeated the exact decomposition. At $h = 2,013$ km, this leave-one-summer-out version gave $+0.40$ percentage points for anomaly energy and -13.67 points for the cross term, compared with $+0.41$ and -13.68 points for the inclusive climatology. The maximum numerical failure of the daily identity was 7.2×10^{-14} . The structural interpretation is therefore insensitive to including the evaluated summer in its climatological reference.

The convergence analysis uses the primary equirectangular distance matrix and compares each grid with the 465-site result using the evaluation events. The 121-site scale profile differs by at most 0.99 percentage points, with a five-scale root mean squared difference of 0.76 points. Latitude refinement reduces these values to 0.50 and 0.38 points; longitude refinement gives 0.93 and 0.73 points. On the 465-site grid, the primary distances and original events give -8.03% ; retaining those distances but reselecting events gives -7.25% . Contraction strengthens with bandwidth at every resolution. Agreement between spatial supports is weakest at 126 km, where the 121-site spacing and station support are coarsest.

Using the primary equirectangular distance matrix, cosine-latitude weighting gave -8.11% with the original labels and -7.67% after relabelling. Replacing the distance matrix by WGS84 geodesic distances gave corresponding estimates of -7.80% and -7.37% ; the WGS84 equal-site estimate was -6.97% . On the full 465-site domain, WGS84 distances with reselected events gave -6.93% , and relabelled boundary checks ranged from -6.32% to -7.97% . The non-rectangular Natural Earth China-land intersection gave -6.35% , with a scale curve of $-2.09, -2.80, -5.06, -9.64$

and -12.18% . Every support retained the negative direction and strengthening contraction with bandwidth, although the magnitude depends on the specified domain.

The reported -7.28% is an equally aggregated relative change on the squared-difference, or semivariance, scale. It corresponds to a -3.71% RMS change when transformed as $\sqrt{1 - 0.0728} - 1$; averaging record-specific RMS ratios gives -3.97% .

Threshold, Stull and sitewise-maximum sensitivities range from -6.90% to -8.27% . Early- and late-period summaries have overlapping descriptive intervals. These checks assess the stability of the spatial contrast across analysis choices. For comparison, the same equal-month, equal-scale and equal-year formula gives -15.14% for the 2015 and 2022 design summers.

5.4. Robustness and external measurement agreement

The 495 middle-day denominators range from 2.25 to $25.99\text{ }^{\circ}\text{C}^2$ across records and scales; within-scale coefficients of variation are 0.076–0.177, with a minimum of $2.25\text{ }^{\circ}\text{C}^2$. These observed denominators do not establish general estimator stability. In the heavy-tail simulation, randomisation calibration coexisted with substantially higher raw-ratio RMSE. The back-transformed log-ratio, our principal robustness result, is -8.30% ; the bounded symmetric contrast $2(Q^H - Q^M)/(Q^H + Q^M)$ is -8.59% . With the primary equirectangular distances, cosine-latitude weighting gives -8.11% using the original labels and -7.67% after relabelling. With WGS84 distances, the corresponding estimates are -7.80% and -7.37% . Kernels matched to the Gaussian weighted mean distance give -7.71% (exponential) and -7.33% (compact quadratic). All preserve the negative direction and strengthening contraction with bandwidth.

The continuous-intensity diagnostic gives a five-scale $\log Q_h$ slope of $-0.065\text{ }^{\circ}\text{C}^{-1}$ (95% descriptive (t)-reference interval based on between-summer variation, $[-0.082, -0.048]$), with negative summer slopes in 31 of 33 years. Scale-specific slopes become more negative from $-0.032\text{ }^{\circ}\text{C}^{-1}$ at 126 km to $-0.110\text{ }^{\circ}\text{C}^{-1}$ at 2,013 km; the corresponding negative-year counts rise from 29 to 32. The negative association extends across the within-month WBT distribution.

The estimated contrast depends on the definition of the spatial field. Subtracting each site's 35-year monthly climatology reduces the estimate to -3.92% ($[-8.67, 0.83]\%$); subtracting the site-year-month mean gives -1.32% ($[-5.63, 2.99]\%$). Site-month standardisation yields -9.87% , but targets a different dimensionless field. Together with the gradient decomposition, these results show that high-day anomalies offset the cross-site climatological gradient, while their own broad-scale energy changes little.

The regime dates have a systematic but nonmonotone seasonal pattern. Averaged over June–August, high days occur 1.96 days later than middle days (95% descriptive (t)-reference interval based on between-summer variation, $[1.10, 2.83]$ days): they occur 7.82 and 6.34 days later in June and July, but 8.29 days earlier in August. Removing a separate linear day-of-month trend within each site-year-month gives -8.81% ($[-13.43, -4.19]\%$); subtracting the leave-one-year calendar-day climatology gives -9.81% ($[-14.22, -5.40]\%$). The negative contrast persists after both seasonal adjustments, although the weaker site-month and site-year-month anomaly results show that different demeaning operations target different spatial fields.

Calendar matching attenuates the contrast but preserves its direction. With 95% descriptive (t)-reference intervals based on between-summer variation in parentheses, the ± 3 -day estimator is -5.21% ($[-8.01, -2.41]\%$), with 24 of 33 summer contrasts negative and 670 of 792 high days matched; the ± 5 -day estimator is -5.72% ($[-8.43, -3.02]\%$), with 25 negative summers and 750 matched high days. Every one of the 99 month-year records has at least one eligible match under both windows. The profiles run from -2.32% to -8.87% for ± 3 days and from -2.54% to -9.89% for ± 5 days, retaining the strengthening contraction with bandwidth.

All fixed-hour and daily-mean analyses yielded negative estimates. Using the original event days, all 24 fixed-hour estimates are negative, from -10.07% at 00 UTC to -6.03% at 11 UTC, and daily-mean fields give -7.58% . When labels are recomputed from each fixed hour, all estimates remain negative, from -14.65% at 22 UTC to -5.42% at 11 UTC; labels recomputed from daily-mean WBT give -9.14% . Peak hours have nearly identical distributions in the original high and middle groups (median 06 UTC). Figure 6 compares the main estimate with transformation, graph-construction, spatial-resolution and field-definition alternatives.

The station comparison contains 175,172 exact-hour matches at 28 usable stations. ERA5-Land minus station WBT has a bias of $-0.263\text{ }^{\circ}\text{C}$, a mean absolute error of $0.938\text{ }^{\circ}\text{C}$ and a root mean squared error of $1.270\text{ }^{\circ}\text{C}$. The

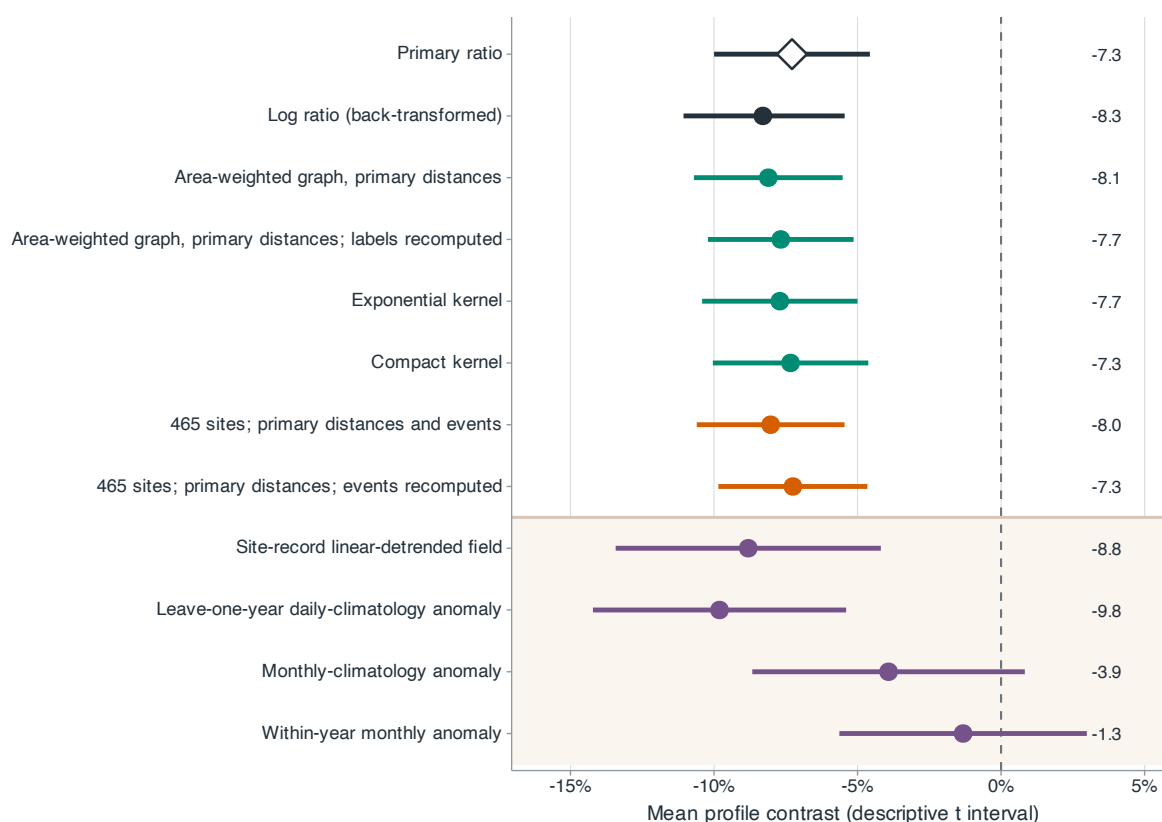


Figure 6 Robustness of the mean profile contrast. Dark rows denote the main and transformation analyses; green, orange and purple rows denote graph construction, spatial resolution and field definition. Points and bars are estimates and descriptive t -reference intervals based on between-summer variation.

within-station centred correlation is 0.916. The two offshore stations excluded from these summaries had no finite values at their nearest ERA5-Land grid cells.

At Gaussian bandwidths 126, 252, 503, 1,006 and 2,013 km, the station high–middle contrasts are -6.23 , -7.23 , -12.50 , -18.03 and -21.47% ; the corresponding ERA5-Land values on identical station subsets are -11.60 , -12.74 , -17.13 , -21.60 and -24.23% . The numbers of negative station summers across the five scales are 6, 7, 8, 9 and 9. Averaged over the three predesignated station-supported scales, the station contrast is -17.34% ($[-28.49, -6.18]\%$), with 9 of 10 summer contrasts negative; the paired ERA5-Land contrast is -20.99% ($[-28.57, -13.40]\%$), with all 10 negative. The station and matched reanalysis estimates agree in direction at the three designated broad scales. At 126 km, both intervals include zero and provide no clear evidence of a local-scale contrast (Figure 7).

Of 240 scheduled high and 440 middle fields, 122 and 223, respectively, had at least 10 stations. The high-minus-middle differences in mean station count and eligible-field fraction were 0.16 sites (descriptive interval $[-2.31, 2.62]$) and 0.002 ($[-0.095, 0.098]$). The three-band station contrast was -18.08% on a fixed common support within each summer. Requiring at least two, three or five retained days per regime gave -13.42 , -15.24 and -14.56% ; the five-day interval included zero because only six summers remained. When regional means over the available station network defined both the daily peak and event labels, the contrast was -15.99% ($[-22.95, -9.02]\%$). Thus the broad direction survives the fixed-support check and a station-based event definition on the available dynamic network, but record sparsity limits precision.

Station elevations ranged from 3 to 2,063 m. In the WGS84 station calculation, treating sea-level pressure as surface pressure instead of applying the elevation conversion changed WBT by 0.072°C on average in absolute value (maximum 1.346°C) and the three-band station contrast from -17.37% to -17.29% .

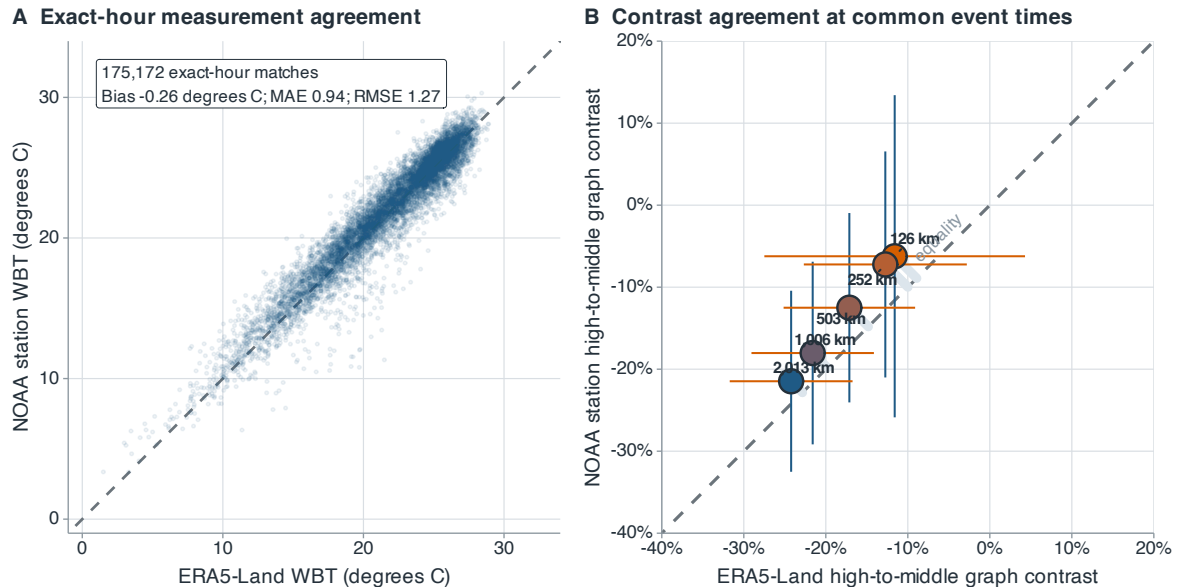


Figure 7 NOAA station and ERA5-Land agreement in ten station-comparison summers. (A) Exact-hour WBT at 28 stations, with summaries from all 175,172 matches. (B) High–middle graph contrasts at the five bandwidths, using ERA5-Land peak times and labels and identical station subsets. Dashed diagonals denote equality. Horizontal and vertical bars in panel B are paired descriptive t -reference intervals based on between-summer variation, not dependence-robust confidence intervals.

6. Discussion

Across the 33 evaluation summers, the weighted squared-difference contrast was 7.28% lower when regional mean WBT was high; this corresponds to a 3.7–4.0% reduction on the RMS pairwise-difference scale. Twenty-eight summer summaries were negative, and the protocol's three-component rule was satisfied. The later product-shift result is exploratory and applies only under the conditional product-invariance null. Contraction strengthened as Gaussian bandwidth increased from $h = 126$ to $h = 2,013$ km, with the largest estimated contraction under domain-wide graph support. The 1950–1990 extension showed the same scale pattern, with a mean of -11.04% and 40 of 41 summer contrasts below zero. Station data also agreed in direction at 503 km and broader: the three-scale mean was -17.34% , with negative contrasts in 9 of 10 summers. At 126 km, the station and matched reanalysis intervals both included zero. These contrasts describe spatial arrangement, a feature distinct from the area exceeding a heat threshold.

Our analysis combines established scale-indexed graph dispersion, finite-record randomisation and exact spatial accounting for an observational field. The five graphs define a single finite-record summary while retaining its scale profile, product shifts preserve within-record dependence among scales, and node and energy decompositions recover the regional contrast exactly. Together, these elements show how the change is distributed over nodes and scales. The finite-record calculation and stochastic process model answer different inferential questions. Product-shift rejection remained near nominal in the simulated null settings. Separate simulations show why Student and HAC intervals can be unreliable with only 33 dependent summers.

The decompositions locate the contrast in the spatial field. After removing each day's regional mean, high-day fields are relatively warmer in the north and cooler in the south than middle-day fields, and the north–south component accounts for most of the contrast. At broad scales, the climatology–anomaly cross component accounts algebraically for nearly all of the contraction, whereas the anomaly field's own energy changes little. High regional mean WBT is therefore associated with a flatter relative geographic pattern. The negative raw-field contrast persists under

area weighting, alternative kernels, fixed-hour analyses and calendar matching. Its attenuation after site-month and site-year-month demeaning further locates the result in the cross-site climatic gradient.

Several limitations remain. The 126-km bandwidth has the coarsest effective spatial support and the weakest station evidence. WGS84 and boundary analyses preserved direction and scale ordering, but estimates varied with the domain; the inference therefore concerns the specified support rather than all of eastern China. The historical extension uses the same reanalysis product, and the 1950–1978 segment carries the documented preliminary-forcing caveat. The station comparison covers 28 locations in ten summers and uses event times and labels from ERA5-Land in the main comparison. Fixed-support and station-defined event checks agree in direction, but stricter day-count requirements reduce the record and widen uncertainty. The comparison measures external consistency, not independent full-field replication. Pressure-level circulation is needed to study a physical mechanism. Population-weighted fields, threshold areas and health data are needed before drawing conclusions about exposure or outcomes. Within the specified domain and record, high-regional-mean days had weaker broad-scale WBT contrast while transient anomaly energy changed little. Generalisation to a long-run climate process requires longer or independent records.

Data availability

ERA5-Land is available from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land>). The NOAA ISD station archive (Smith et al., 2011) is available at <https://www.ncei.noaa.gov/products/land-based-station/integrated-surface-database>. The processed data, site and station manifests, quality-control records, derived results and complete analysis code are available in the public GitHub repository <https://github.com/Stork343/eastern-china-wet-bulb-contraction>. The submission-matched code, protocols, results and PDFs are frozen in the release <https://github.com/Stork343/eastern-china-wet-bulb-contraction/releases/tag/jrssc-submission-v1-2026-08-13>. The raw reanalysis and station archives remain available through their source providers.

Acknowledgements

OpenAI Codex was used for language checking and code debugging. The authors reviewed all changes and take full responsibility for the manuscript and analysis.

Funding

This work was supported by the Beijing Natural Science Foundation, “Theory, Methodology and Applications of Functional Hierarchical Quantile Regression Modeling” (grant 1242005); the Fundamental Research Funds for the Central Universities and the Research Funds of Renmin University of China, “Robust Statistical Inference for Complex Data” (grant 25XNN015); and the Ministry of Education Humanities and Social Sciences Research General Project, “Research on Spatial-temporal Quantile Regression Modeling: Theoretical Methods and Applications” (grant 25YJA910005).

Conflict of interest

The authors declare no conflict of interest.

References

- Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59, 817–858.
- Bolton, D. (1980). The computation of equivalent potential temperature. *Monthly Weather Review*, 108, 1046–1053.
- Chagnaud, G., Taylor, C.M., Jackson, L.S., Birch, C.E., Marsham, J.H. and Klein, C. (2025). Wet-bulb temperature extremes locally amplified by wet soils. *Geophysical Research Letters*, 52, e2024GL112467.
- Cressie, N.A.C. (1993). *Statistics for Spatial Data*, revised edn. New York: Wiley.
- Davies-Jones, R. (2008). An efficient and accurate method for computing the wet-bulb temperature along pseudoadiabats. *Monthly Weather Review*, 136, 2764–2785.

- Dong, X., Thanou, D., Frossard, P. and Vandergheynst, P. (2016). Learning Laplacian matrix in smooth graph signal representations. *IEEE Transactions on Signal Processing*, 64, 6160–6173.
- Geary, R.C. (1954). The contiguity ratio and statistical mapping. *The Incorporated Statistician*, 5, 115–145.
- García-Soidán, P.H., Febrero-Bande, M. and González-Manteiga, W. (2004). Nonparametric kernel estimation of an isotropic variogram. *Journal of Statistical Planning and Inference*, 121, 65–92. doi:10.1016/S0378-3758(02)00507-4.
- Healy, D., Tawn, J., Thorne, P. and Parnell, A. (2025). Inference for extreme spatial temperature events in a changing climate with application to Ireland. *Journal of the Royal Statistical Society: Series C*, 74, 275–299. doi:10.1093/jrsssc/qlae047.
- Harris, K.D. (2020). A shift test for independence in generic time series. *arXiv:2012.06862*.
- Horton, D.E., Johnson, N.C., Singh, D., Swain, D.L., Rajaratnam, B. and Diffenbaugh, N.S. (2015). Contribution of changes in atmospheric circulation patterns to extreme temperature trends. *Nature*, 522, 465–469.
- Ibragimov, I.A. and Linnik, Y.V. (1971). *Independent and Stationary Sequences of Random Variables*. Groningen: Wolters-Noordhoff.
- Moran, P.A.P. (1950). Notes on continuous stochastic phenomena. *Biometrika*, 37, 17–23.
- Muñoz-Sabater, J. et al. (2021). ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13, 4349–4383.
- Newey, W.K. and West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55, 703–708.
- Pepin, N.C. et al. (2015). Elevation-dependent warming in mountain regions of the world. *Nature Climate Change*, 5, 424–430.
- Phipson, B. and Smyth, G.K. (2010). Permutation p -values should never be zero: calculating exact p -values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9, Article 39.
- Raymond, C., Matthews, T. and Horton, R.M. (2020). The emergence of heat and humidity too severe for human tolerance. *Science Advances*, 6, eaaw1838.
- Seneviratne, S.I., Corti, T., Davin, E.L., Hirschi, M., Jaeger, E.B., Lehner, I., Orlowsky, B. and Teuling, A.J. (2010). Investigating soil moisture–climate interactions in a changing climate: a review. *Earth-Science Reviews*, 99, 125–161.
- Sherwood, S.C. and Huber, M. (2010). An adaptability limit to climate change due to heat stress. *Proceedings of the National Academy of Sciences*, 107, 9552–9555.
- Shreevastava, A., Raymond, C. and Hulley, G. (2023). Contrasting intraurban signatures of humid and dry heatwaves over Southern California. *Journal of Applied Meteorology and Climatology*, 62, 709–720.
- Shuman, D.I., Narang, S.K., Frossard, P., Ortega, A. and Vandergheynst, P. (2013). The emerging field of signal processing on graphs: extending high-dimensional data analysis to networks and other irregular domains. *IEEE Signal Processing Magazine*, 30, 83–98.
- Simmons, J.P., Nelson, L.D. and Simonsohn, U. (2011). False-positive psychology: undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22, 1359–1366.
- Smith, A., Lott, N. and Vose, R. (2011). The integrated surface database: recent developments and partnerships. *Bulletin of the American Meteorological Society*, 92, 704–708. doi:10.1175/2011BAMS3015.1.
- Speizer, S., Raymond, C., Ivanovich, C. and Horton, R.M. (2022). Concentrated and intensifying humid heat extremes in the IPCC AR6 regions. *Geophysical Research Letters*, 49, e2021GL097261.
- Stull, R. (2011). Wet-bulb temperature from relative humidity and air temperature. *Journal of Applied Meteorology and Climatology*, 50, 2267–2269.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68, 1097–1126.
- Wood, S.N. (2003). Thin plate regression splines. *Journal of the Royal Statistical Society: Series B*, 65, 95–114.
- Yuan, A.E. and Shou, W. (2024). A rigorous and versatile statistical test for correlations between stationary time series. *PLOS Biology*, 22, e3002758.

Table 1. Boundary of the statistical contribution.

Component	Established basis	Contribution in this study
Graph Dirichlet energy	Graph-signal roughness	Relative spatial comparison between regional-mean states
Kernel-weighted semivariance	Distance-weighted variogram summaries	One hierarchical target across five prespecified bandwidths
Cyclic shifts	Group-based randomisation tests	Record-wise joint shifts of complete multiscale daily profiles
Node allocation	Edge-contribution splitting	Additivity matched to the relative contrast and aggregation hierarchy
Laplacian projection	Graph-orthogonal decomposition	Additive structural localisation of state contrasts
Application insight	Not a statistical primitive	Association between high regional humid heat and contraction of the north-south WBT contrast

Table 2. Analysis status, timing and reconciliation with the Version 2 protocol.

Analysis	Original protocol status	Fixed before the 33-summer result?	Final manuscript role	Reason for final role
33-summer, five-scale mean	Primary estimand	Yes	Primary finite-record summary	Protocol specification retained
Student mean test	Confirmatory component	Yes	Protocol result and process reference	Dependence simulations show under-coverage
Lag-2 HAC mean test	Confirmatory component	Yes	Protocol result and process reference	A fixed lag is not generally consistent
Independent-summer sign reference	Confirmatory component	Yes	Protocol recurrence reference	Exact binomial interpretation requires independent summer signs
Three-component rule	Confirmatory decision rule	Yes	Transparently reported protocol decision	Rule was satisfied
Global product shift	Not prespecified	No	Exploratory conditional randomisation	Added after the primary estimate
1950–1990 analysis	Pre-access extension	Before historical data access	Temporal extension	Extension specification retained
NOAA comparison	Pre-access extension	Before station-result access	External measurement comparison	Uses reanalysis-defined events

Table 3. Physical scale of the evaluation-period graph contrasts. h is the Gaussian bandwidth, not a pair-distance cutoff. $\bar{d}_w$ is the edge-weighted mean pair distance, and N_{eff} is the Kish effective edge count. Graph contrasts are relative changes in weighted squared differences. RMS contrasts average record-specific ratios of $(2Q)^{1/2}$; Negative is the number of negative summer graph contrasts.

h (km)	$\bar{d}_w$ (km)	N_{eff} (edges)	Graph contrast (%)	RMS contrast (%)	Negative (/33)
126	200	366	−2.86	−1.54	22
252	319	1,138	−3.94	−2.14	23
503	549	3,223	−5.91	−3.25	25
1,006	826	6,005	−10.44	−5.70	29
2,013	992	7,109	−13.27	−7.22	30